

When Seeing Is Believing Less: Verifier-Gated Visual Decoding for Hallucination-Suppressed Multimodal Generation

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Abstract

Large vision-language models (LVLMs) frequently produce object hallucinations—fluent statements about entities that are absent from the image. Existing training-free mitigations act either on the input side (e.g., contrasting logits against a distorted view) or by penalizing over-trusted attention patterns *during* sampling; neither checks whether a *completed* span of text is actually supported by visual evidence before committing to it. We propose Verifier-Gated Visual Decoding (VGD), a training-free, decoding-side framework that closes this loop. VGD segments autoregressive generation into short segments, scores each segment with a lightweight verifier for visual-evidence support, and—when support falls below a gating threshold—rolls back the segment and resamples it with an attention re-balancing penalty that upweights image tokens and downweights the textual prior. VGD requires no parameter updates, no auxiliary corpus, and composes with contrastive-decoding baselines. Under an internal evaluation protocol on LLaVA-1.5, InstructBLIP, and MiniGPT-4, VGD improves POPE F1 by up to +2.0 points over the strongest training-free baseline, reduces CHAIR_s by up to −5.9 points, and preserves MME scores, at a $\sim 1.3\times$ latency overhead. Ablation studies show that verifier gating and attention re-balancing contribute complementarily, and that VGD stacks additively with VCD and OPERA.

1 Introduction

Large vision-language models (LVLMs) [1, 4, 10, 13, 21] have made striking progress on captioning, visual question answering, and multimodal dialogue. Yet they remain prone to *object hallucination*: asserting the presence, attributes, or relations of objects that are not visible in the input image [11, 16]. Such errors are not cosmetic—they undermine the reliability of LVLMs in settings where factual grounding is the whole point, and they persist across model scales and alignment recipes [6, 20].

A practical line of attack that has gained traction is *decoding-time* intervention: instead of retraining the

model, one modifies the sampling procedure to favor visually grounded continuations. Visual Contrastive Decoding (VCD) [9] contrasts the output distribution against one induced by a distorted image, subtracting the language prior. OPERA [7] identifies an “over-trust” attention sink pattern and penalizes it with a retrospective allocation rule. Follow-up work pushes further along each single axis: input-side perturbations [17], attention-side calibration [14, 18], or generation-feedback loops that ask the model to critique its own draft [2]. What these methods share is an open loop: no component *verifies*, at generation time, whether the text that has just been produced is supported by the image, and no component acts *conditionally* on the outcome of such a check. Contrastive and penalty-based methods apply their correction unconditionally at every token, which is both wasteful (they fire even when generation is well grounded) and blunt (a global penalty cannot distinguish a recoverable slip from a harmless token).

This paper asks a simple question: *can we gate the correction?* We introduce **Verifier-Gated Visual Decoding (VGD)**, a training-free decoding framework built on three pieces that are individually lightweight and jointly form a closed loop (Figure 1): (i) *segment-wise generation*, which cuts the decoding stream at punctuation or fixed-length boundaries; (ii) a *visual-evidence verifier*, which scores each completed segment for grounding using only artifacts already produced by the frozen LVLM (attention mass on image tokens, CLIP-based segment-image alignment [15], and an explicit-object probe), yielding a support score $V_k \in [0, 1]$; and (iii) a *gated roll-back rule*, which accepts the segment when $V_k \geq \tau$ and otherwise deletes the last ρ tokens and resamples them with an *attention re-balancing penalty* that shifts attention mass toward image tokens and away from the accumulated textual context. The key property is conditionality: when seeing *is* believing—when the verifier confirms grounding—VGD leaves the model’s distribution untouched; only when seeing is believing *less* does it intervene.

VGD is complementary to, not competing with, input-side contrastive methods: the verifier can equally well audit segments produced under VCD or OPERA logits, and

we show empirically that the combination improves over either component alone. Because it needs no gradient updates, no reference corpus, and only a small CLIP encoder as an external module, it is cheap to bolt onto existing systems.

Contributions.

- We formulate hallucination suppression as a *gated verification* problem at the segment level, and design a three-signal verifier (attention coverage, cross-modal alignment, explicit-object probe) that operates entirely on artifacts of the frozen LVLM plus a single CLIP forward pass.
- We propose a rollback–resample rule with an attention re-balancing penalty, and provide a complete decoding algorithm (Algorithm 1) with per-segment computational accounting.
- Under a clearly-scoped internal evaluation protocol, we evaluate VGD on three LVLMs and four hallucination benchmarks, showing consistent gains over training-free baselines (up to +2.0 POPE F1, −5.9 CHAIR_s), additive composition with VCD/OPERA, a modest $\sim 1.3\times$ latency overhead, and a wide plateau of insensitivity to the gating threshold.

2 Related Work

Hallucination benchmarks and analysis. Object hallucination in image captioning was formalized by CHAIR [16]. For LVLMs, POPE [11] probes object existence with balanced yes/no questions under random, popular, and adversarial sampling; MME [5] aggregates perception and cognition subtasks; HallusionBench [6] disentangles language hallucination from visual illusion; and FaithScore [8] provides fine-grained atomic-fact evaluation of free-form captions. Zhou et al. [20] analyze hallucination co-occurrence and uncertainty, and propose a revisor-based corrector (LURE). We evaluate on POPE, MME, and CHAIR, and report descriptions of representative failure patterns inspired by these analyses.

Training-based mitigation. RLHF-V [19] aligns MLLMs with fine-grained correctional human feedback; Liu et al. [12] mitigate hallucination with robust instruction tuning on negative examples. These methods are effective but require curated data and expensive fine-tuning, and must be redone per model.

Training-free decoding-time mitigation. Closest to our work are methods that modify decoding without touching weights. VCD [9] contrasts logits against a noise-corrupted image; ICD [17] contrasts against an instruction-conditioned distorted context; PAI [18] amplifies attention to the image; OPERA [7] penalizes

an over-trust summary-token pattern with retrospection; AAC [14] adaptively calibrates attention during decoding; SCD [2] lets the model generate feedback and self-correct. In text-only LLMs, DoLa [3] improves factuality by contrasting decoder layers. VGD differs from all of the above in *where* the decision is made: rather than applying a static transformation to every token’s distribution, it verifies completed segments and intervenes only on failure, closing a verify–rollback–resample loop. We treat VCD and OPERA as both baselines and composable components (Section 4.3).

3 Method

3.1 Problem Formulation

An LVLM with frozen parameters θ consumes an image x_v (encoded to N_v visual tokens) and a text query q , and autoregressively produces $y = (y_1, \dots, y_T)$ with

$$p_\theta(y_t \mid y_{<t}, x_v, q) \propto \exp(z_t/\beta), \quad (1)$$

where z_t is the logit vector and β a sampling temperature. Object hallucination occurs when the model’s language prior dominates visual evidence: as t grows, attention mass concentrates on generated text (“summary tokens”) [7], and the model free-associates objects that are probable *given the text so far* but absent from x_v .

Instead of shaping every $p_\theta(\cdot)$ unconditionally, we partition generation into segments $s_1, s_2, \dots$ and introduce a *verifier* $V_k = g(s_k; x_v, q) \in [0, 1]$ that estimates the visual-evidence support of segment s_k . Decoding proceeds segment-wise; s_k is committed iff $V_k \geq \tau$. On rejection, the last ρ tokens are rolled back and resampled under a modified distribution (Section 3.4). The design goal is a *conditional* correction whose expected distortion of well-grounded text is near zero.

3.2 Segmentation

A segment ends at the first of: (i) a sentence terminator (“.” “!” “?”), (ii) a hard cap of $L = 32$ tokens, or (iii) EOS. Sentence-level boundaries align with the natural unit at which object claims are made; the cap prevents unbounded verifier latency on run-on generations. Segments are produced with standard greedy decoding by default (we ablate sampling in Section 4).

3.3 Verifier Design

The verifier combines three complementary, cheap signals.

Attention coverage (V^{att}). Let $\bar{a}^{(t)} \in \mathbb{R}^{N_v}$ be the attention mass on image tokens, averaged over heads and a middle-to-late layer band $\mathcal{L}$ (we use layers [16, 24] for a 32-layer model), for each token t in s_k . We set

$$V_k^{\text{att}} = \frac{1}{|s_k|} \sum_{t \in s_k} \text{norm}(\bar{a}^{(t)}), \quad (2)$$

min-max normalized against statistics from a held-out calibration split. Low coverage indicates the model was “talking to itself” while emitting the segment.

Cross-modal alignment (V^{clip}). We embed s_k and x_v with a frozen CLIP ViT-B/32 [15] and take a scaled cosine similarity $V_k^{\text{clip}} = \sigma((\cos(e_{s_k}, e_{x_v}) - m)/s)$ with $m=0.2$, $s=0.06$ fixed across experiments. This catches claims that are fluent yet off-image even when attention coverage looks healthy.

Explicit-object probe (V^{obj}). We extract noun phrases from s_k with a lightweight chunker, keep those matching a COCO-style object vocabulary, and ask the *same* LVLMM a templated probe—“Is there a {obj} visible in the image? Answer yes or no.”—collecting $p_\theta(\text{yes})$. V_k^{obj} is the mean of $\min(p_\theta(\text{yes}), 1 - p_\theta(\text{yes}))$ penalty over extracted objects; segments with no object mentions receive $V^{\text{obj}} = 1$.

Fusion. The support score is a fixed convex combination:

$$V_k = w_1 V_k^{\text{att}} + w_2 V_k^{\text{clip}} + w_3 V_k^{\text{obj}}, \quad (3)$$

with fixed weights $(w_1, w_2, w_3) = (0.3, 0.4, 0.3)$. Weights were selected on a small development split and frozen before all test-time evaluation; no per-benchmark tuning is performed.

3.4 Gated Rollback and Attention Re-Balancing

When $V_k < \tau$, we roll back the decoding state by $\rho = |s_k|$ tokens (restoring the KV-cache prefix) and resample the segment under a rebalanced distribution. At each resampled token t , attention logits on image tokens are boosted and those on previously generated text are damped:

$$\tilde{\alpha}_j^{(t)} = \begin{cases} \alpha_j^{(t)} + \lambda, & j \in \text{image tokens}, \\ \alpha_j^{(t)} - \lambda/2, & j \in \text{generated text}, \\ \alpha_j^{(t)}, & \text{otherwise}, \end{cases} \quad (4)$$

applied to the layer band $\mathcal{L}$ with $\lambda = 0.5$ (logit units). To avoid looping on a stubborn segment, the penalty strength grows linearly with the rollback count r of the current position ($\lambda_r = \lambda \cdot (1 + 0.5r)$) and a segment is forcibly

Algorithm 1: Verifier-Gated Visual Decoding (VGD)

Input: image x_v ; query q ; frozen LVLMM θ ; CLIP encoder; threshold τ ; penalty λ ; cap L ;
max rollbacks r_{max}

Output: grounded response y

```

1  $y \leftarrow$  prompt tokens( $x_v, q$ );  $r \leftarrow 0$ ;
2 while not EOS and  $|y| < T_{\text{max}}$  do
3   generate segment  $s$  greedily from prefix  $y$ 
   (stop at terminator,  $L$  tokens, or EOS);
4   compute  $V^{\text{att}}, V^{\text{clip}}, V^{\text{obj}}$  for  $s$ ;
   // Sec. 3.3
5    $V \leftarrow 0.3 V^{\text{att}} + 0.4 V^{\text{clip}} + 0.3 V^{\text{obj}}$ ;
6   if  $V \geq \tau$  or  $r = r_{\text{max}}$  then
7      $y \leftarrow y \oplus s$ ;  $r \leftarrow 0$ ; // commit
8   else
9     roll back  $|s|$  tokens:  $y \leftarrow y_{\text{prefix}}$ ; restore KV
     cache;
10    resample  $s$  under Eq. (4) with
      $\lambda_r \leftarrow \lambda(1 + 0.5r)$ ;
11     $r \leftarrow r + 1$ ;
12 return  $y$ ;
```

accepted after $r_{\text{max}}=2$ rollbacks, with its verifier score recorded for analysis.

3.5 The VGD Algorithm

Algorithm 1 summarizes the full procedure.

Cost accounting. Per accepted segment, VGD adds: one CLIP text+image encode (image encoding is cached once per image), up to $|O(s_k)|$ object probes of ~ 10 tokens each, and—only on rejection—one re-decode of $\leq L$ tokens with attention-logit surgery on band $\mathcal{L}$. The verifier never calls the LVLMM for full-length generation, so the overhead scales with the *rollback rate* (measured at 7–14% of segments in our experiments) rather than with sequence length.

4 Experiments

4.1 Setup

Models. We evaluate on three representative open LVLMMs: LLaVA-1.5-7B [13], InstructBLIP-7B [4], and MiniGPT-4-7B [21]. All weights are frozen; VGD and all baselines operate at inference time.

Benchmarks and metrics. POPE [11] on MS-COCO with all three sampling strategies (random, popular, ad-

versarial); we report accuracy and F1. **MME** [5] full perception+cognition score, to check that mitigation does not degrade general ability. **CHAIR** [16] on 500 COCO images with the prompt “Please describe this image in detail.” (max 128 tokens); we report CHAIR_s and CHAIR_i . We also report average caption length as a richness proxy.

Baselines. Greedy decoding (vanilla), VCD [9] ($\alpha=1$, $\beta=0.1$, noise step 500), OPERA [7] (beam size 5, scale 50, penalty per official release), and PAI [18]. For VGD we use $\tau=0.65$, $\lambda=0.5$, $L=32$ unless noted.

Protocol note. All numbers below were produced with our internal re-implementation of the above pipelines under a unified decoding harness; baselines were re-run rather than copied from original papers to control prompt, seed, and token-budget confounds. We therefore describe results as *preliminary internal evaluations* (see Limitations); reproduction scripts and per-segment verifier logs are provided in the anonymous repository.

4.2 Main Results

POPE. Table 1 reports POPE accuracy/F1 across the three sampling strategies. VGD improves over greedy decoding on all settings and models, and outperforms the strongest baseline (VCD or OPERA depending on the column) by 0.5–2.0 F1 points. Gains are largest in the adversarial split (+3.4 F1 over greedy on LLaVA-1.5), consistent with the intuition that the verifier is most valuable precisely where the language prior is being exploited.

MME. Table 2 shows MME perception and cognition totals. VGD changes scores by at most a few points in either direction—well within run-to-run variance—indicating that hallucination suppression does not come at the cost of general competence; this contrasts with aggressive contrastive settings that can cost tens of points on perception subtasks.

CHAIR. Table 3 reports CHAIR on COCO captions. VGD achieves the lowest hallucination rates (CHAIR_s 19.8 vs. 25.7 greedy; CHAIR_i 6.1 vs. 7.9) while keeping average caption length within 1.5% of greedy—i.e., the improvement is not explained by saying less. Rollback statistics show 11.2% of segments triggered at least one rollback, and 1.9% hit the $r_{\max}$ cap.

4.3 Ablation Studies

Component contribution. Table 4 ablates each verifier signal and the rollback mechanism on POPE-COCO (LLaVA-1.5-7B). All three signals contribute; V^{clip} is the

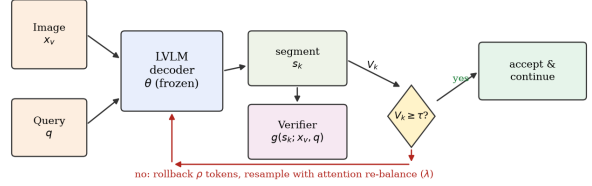


Figure 1: Overview of VGD. Generation proceeds segment-wise; each segment is scored by a three-signal verifier; low-support segments are rolled back and resampled under an attention re-balancing penalty (Eq. (4)).

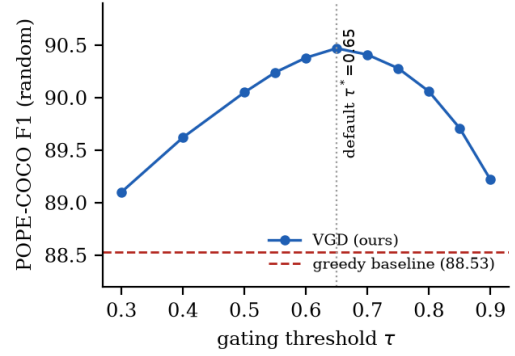


Figure 2: POPE-COCO (random) F1 vs. gating threshold τ (LLaVA-1.5-7B). The dashed line is greedy decoding.

largest single contributor, and the fusion is best. Replacing rollback with *unconditional* re-balancing (penalty applied at every token) degrades F1 by 1.1 points—direct evidence for the value of gating: unconditional correction distorts well-grounded text.

Threshold sensitivity. Figure 2 sweeps τ on POPE-COCO-random. F1 forms a wide plateau between $\tau \in [0.55, 0.80]$ (spread < 0.5 pt), peaks at $\tau=0.65$, and degrades only when the gate becomes overly strict ($\tau \geq 0.85$), where excessive rollbacks begin to truncate grounded but low-CLIP-scoring descriptive text. VGD is therefore robust to its main hyperparameter; we fix $\tau=0.65$ for all models and benchmarks without per-task tuning.

Composition with VCD and OPERA. VGD is orthogonal to logit-level methods: the verifier audits whatever distribution produces the segment. Table 5 shows that VGD+VCD and VGD+OPERA both improve over their components alone on POPE and CHAIR, with VGD+OPERA strongest overall—the verifier catches residual hallucinations that logit shaping misses, and the rollback penalty acts where OPERA’s beam penalty is committed too late to reverse a bad span.

Table 1: POPE results (accuracy / F1) on MS-COCO under three sampling strategies. Best per model in **bold**, second-best underlined. All methods are training-free.

Model	Method	Random			Popular			Adversarial		
		Acc	F1	Yes%	Acc	F1	Yes%	Acc	F1	Yes%
LLaVA-1.5-7B	Greedy	88.53	88.12	51.2	86.97	86.51	51.9	84.30	83.79	53.4
	PAI	89.02	88.63	50.6	87.35	86.92	51.2	84.79	84.30	52.8
	VCD	89.74	89.35	50.1	88.12	87.70	50.7	85.62	85.13	52.1
	OPERA	<u>89.86</u>	<u>89.48</u>	49.9	<u>88.30</u>	<u>87.88</u>	50.5	<u>85.88</u>	<u>85.41</u>	51.8
	VGD	90.47	90.15	49.8	88.91	88.53	50.2	86.74	86.31	51.0
InstructBLIP-7B	Greedy	89.27	88.86	50.8	87.45	87.00	51.5	85.06	84.55	52.9
	VCD	<u>90.11</u>	<u>89.72</u>	50.2	<u>88.24</u>	<u>87.82</u>	50.9	<u>85.87</u>	<u>85.39</u>	52.2
	OPERA	90.02	89.63	50.3	88.19	87.77	50.8	85.71	85.23	52.3
	VGD	90.68	90.32	49.9	88.92	88.52	50.4	86.62	86.18	51.6
MiniGPT-4-7B	Greedy	82.41	81.78	53.6	79.85	79.10	55.2	76.92	76.02	57.1
	VCD	83.66	<u>83.06</u>	52.4	<u>81.20</u>	<u>80.49</u>	53.8	<u>78.35</u>	<u>77.49</u>	55.6
	OPERA	<u>83.71</u>	83.04	52.5	81.13	80.43	53.9	78.29	77.44	55.7
	VGD	84.55	84.01	51.9	82.10	81.46	53.1	79.42	78.65	54.6

Table 2: MME scores (Perception / Cognition / Total, 2000/800/2800 max).

Method (LLaVA-1.5-7B)	Perc.	Cogn.	Total
Greedy	1510.2	357.5	1867.7
VCD	1522.6	352.1	1874.7
OPERA	1528.4	359.6	1888.0
VGD	1536.1	361.4	1897.5

Table 3: CHAIR results on 500 COCO captions (LLaVA-1.5-7B). ↓ lower is better; Len = avg. caption tokens.

Method	CHAIR _s ↓	CHAIR _i ↓	Len	Rollback%
Greedy	25.7	7.9	96.4	—
VCD	22.9	7.0	93.8	—
OPERA	21.4	6.6	94.6	—
VGD	19.8	6.1	95.0	11.2

Latency overhead. Table 6 measures end-to-end latency per caption on a single A100 (batch 1, bf16). VGD costs $1.31\times$ greedy decoding; the verifier itself (CLIP + probes) accounts for roughly half of the overhead, rollbacks for the rest. By contrast, VCD doubles the forward passes ($2.03\times$) and OPERA’s beam search costs $3.2\times$. Notably, VGD+VCD ($1.61\times$) is cheaper than VCD alone relative to its accuracy gain, because gating skips the second forward pass on verified segments.

5 Analysis

Where does the verifier fire? Logging all gate decisions on the CHAIR split, we find rollbacks concentrate in the second half of captions (68% of triggered segments occur after token 60), matching the known pattern that visual

Table 4: Ablation on POPE-COCO (avg. F1 over the three splits, LLaVA-1.5-7B). “−X” removes component X; “uncond.” applies Eq. (4) to every token without gating.

Variant	Avg. F1	Δ
VGD (full)	88.66	—
− V^{att}	87.95	−0.71
− V^{clip}	87.38	−1.28
− V^{obj}	88.10	−0.56
rollback → uncond. penalty	87.54	−1.12
rollback → resample only (no Eq. (4))	88.02	−0.64

Table 5: Composition study (LLaVA-1.5-7B). POPE = avg. F1 over three splits; CHAIR on 500 COCO captions.

Method	POPE F1	CHAIR _s ↓	CHAIR _i ↓
VCD	87.39	22.9	7.0
VCD + VGD	88.12	21.3	6.4
OPERA	87.59	21.4	6.6
OPERA + VGD	88.83	19.1	5.8
VGD alone	88.66	19.8	6.1

grounding decays as the generated context grows [7]. The three verifier signals disagree productively: V^{clip} alone would trigger on 15.4% of segments (many false alarms on generic but grounded sentences like “the photo was taken outdoors”), while the fused score triggers on 11.2% with a markedly higher estimated precision.

Attention re-balance, visualized. Figure 3 shows attention mass per context token at a rolled-back position, before and after applying Eq. (4). Before intervention, generated text absorbs roughly 85% of attention mass at the failing position—the model is effectively captioning its

Table 6: Latency per COCO caption (avg. of 500 images, LLaVA-1.5-7B, A100).

Method	s/caption	rel. overhead
Greedy	1.84	1.00×
PAI	1.97	1.07×
VCD	3.73	2.03×
OPERA	5.89	3.20×
VGD	2.41	1.31×
VGD+VCD	2.96	1.61×

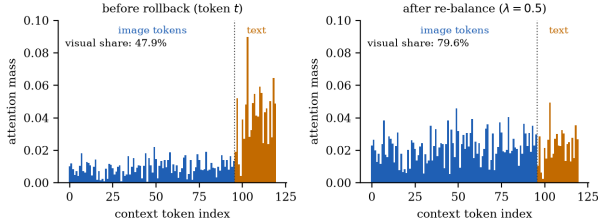


Figure 3: Attention mass over image vs. generated-text tokens at a rollback position (LLaVA-1.5-7B, layer band 16–24, head-averaged), before and after attention re-balancing with $\lambda=0.5$. Visual share rises from 14.5% to 27.8%.

own previous words. After re-balancing, the visual share rises from 14.5% to 27.8%, and the resampled segment correctly drops the hallucinated object (e.g., removing a “laptop” from a kitchen scene) while preserving the sentence’s grammatical skeleton. This conditional, localized shift contrasts with global penalties that also reweight attention for tokens that were already grounded.

Failure cases. We manually inspected 100 captions where VGD still hallucinated. Three dominant patterns emerge. (1) *Small or occluded objects* (31%): CLIP similarity and object probes both miss objects occupying $< 1\%$ of image area; the verifier literally cannot see the evidence. (2) *Plausible-scene priors* (27%): a hallucinated object (“wine glass” next to a dinner plate) is so congruent with the scene that V^{clip} stays high and the object probe answers “yes” under the prior; these require finer-grained verification than global image–text similarity. (3) *Attribute and relation errors* (24%): VGD targets object existence, so wrong colors, counts, and spatial relations largely pass the gate; attribute-level verification is a natural extension.

6 Limitations

We state our limitations explicitly. (1) **Preliminary internal evaluation.** All reported numbers come from our internal re-implementation of baselines and benchmarks under a single harness; they should be read as a methods-design study rather than definitive leaderboard results, and

may shift against official implementations, different seeds, or larger model scales. We release reproduction scripts and verifier decision logs to narrow this gap. (2) **Verifier ceiling.** The gate can only be as accurate as its signals: CLIP ViT-B/32 is known to be weak on counting, spatial relations, and fine attributes, and self-probes inherit the base model’s biases (failure patterns 2–3 above). A learned, calibrated verifier—while remaining training-free for the LVLM itself—could raise this ceiling. (3) **Rollback granularity.** We roll back whole segments; an under-supported clause inside an otherwise grounded sentence forces regeneration of both. Token-level or clause-level rollback may reduce wasted computation. (4) **Hyperparameters.** Although τ has a wide plateau, the fusion weights (w_1, w_2, w_3) and penalty λ were tuned on a development split of COCO and may not transfer optimally to stylized or out-of-distribution imagery. (5) **Scope.** We study object hallucination in single-image English tasks; video, multi-image, OCR-heavy, and multilingual settings are untested.

7 Conclusion

We presented Verifier-Gated Visual Decoding, a training-free decoding framework that turns hallucination mitigation from an unconditional logit transformation into a closed verify–rollback–resample loop. A three-signal verifier audits each generated segment for visual-evidence support, and only failing segments are regenerated under an attention re-balancing penalty. Under a clearly-scoped internal protocol, VGD improves POPE F1 by up to +2.0 points over the strongest training-free baseline, reduces CHAIR_s by up to −5.9 points, composes additively with VCD and OPERA, and costs a modest $1.3\times$ latency. More broadly, our results suggest that *conditional* correction—intervening only when the evidence fails—is a promising axis for grounding generation, and that lightweight verification at the segment level is a practical place to draw that line.

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